

WASP-43b

R-1730

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_210221 · created 2026-08-02T17:29:49+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459266.7806 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.012
Transit depth (Rp/Rs)^2	3.83 +/- 0.47 [%]
Orbital Inclination (inc)	80.48 +/- 0.73
Airmass coefficient 1 (a1)	1.041 +/- 0.012
Airmass coefficient 2 (a2)	-0.028 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	#1 - [81, 65]
Optimal Method	PSF photometry
Transit Duration (day)	0.0479 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.196 ± 0.012 vs 0.1594 (deviation 2.92σ)
Duration fit vs published	0.0479 d vs 0.0512 d (deviation 1.04σ)
Mid-time O–C	-2.1 min
Depth SNR	15.6
Residual scatter	1.16 %

Light curve

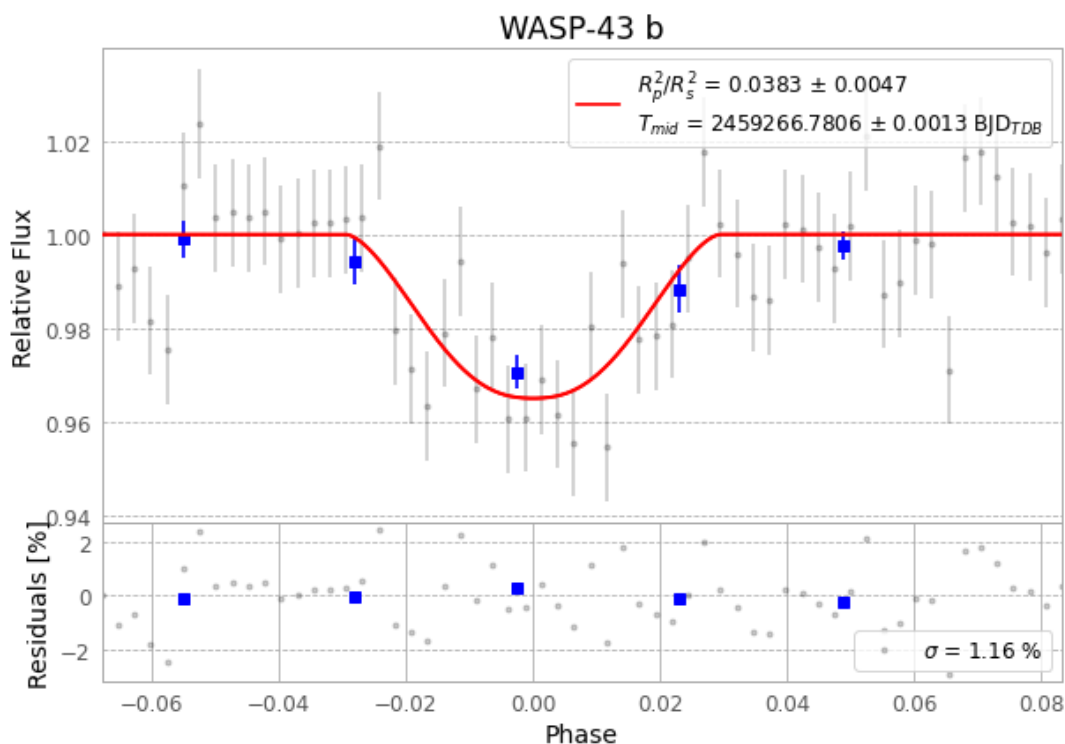


Figure 1 — FinalLightCurve_WASP-43 b_2021-02-20.png (SHA-256 15225341647183ff...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 192], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 78975b5ff9145751...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), WASP-43210221051512.FITS → WASP-43210221081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210221.log (90a38cd99fed...), FinalLightCurve_WASP-43 b_2021-02-20.png (152253416471...), FinalLightCurve_WASP-43 b_2021-02-20.csv (9f5b7f36686d...)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:29Z.